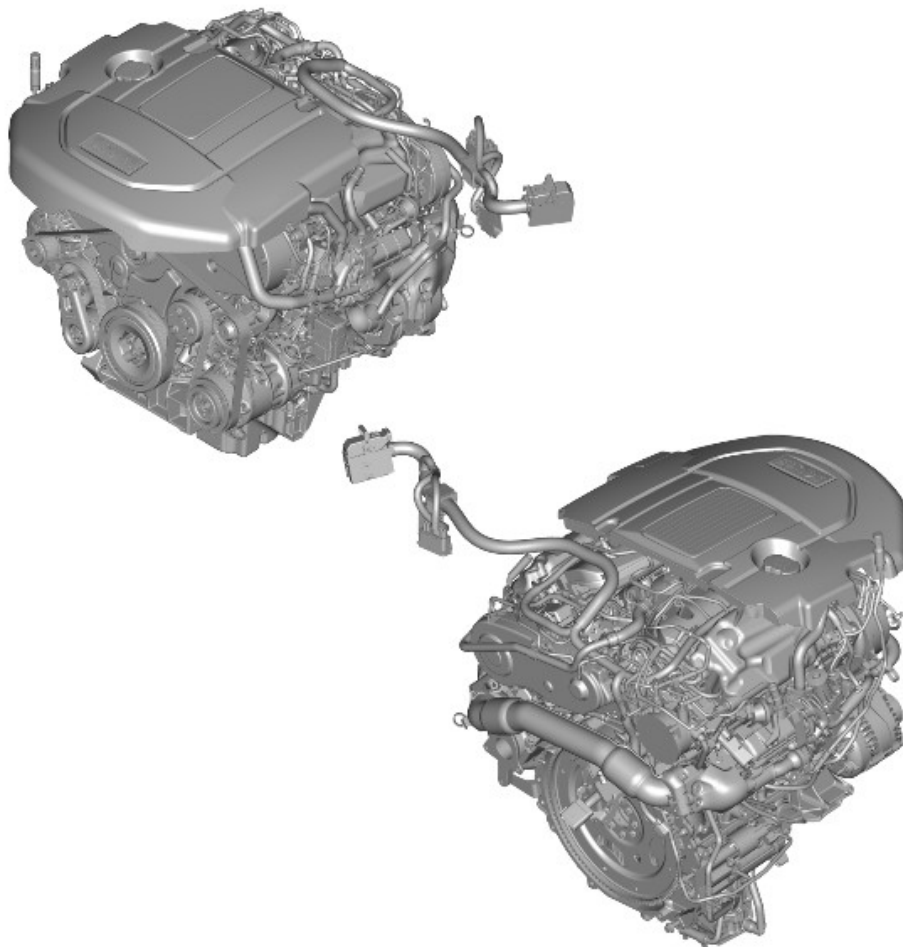


2016.0 RANGE ROVER (LG), 303-01

ENGINE – TDV6 3.0L DIESEL – GEN 2/TDV6 3.0L DIESEL

DESCRIPTION AND OPERATION

EXTERNAL VIEWS



E169818

OVERVIEW

The 3.0 liter diesel engine is a V6 configuration unit with 2 banks of 3 cylinders arranged at 60 degrees to each other. There are 4 valves per cylinder, which are operated by 2 overhead camshafts per cylinder bank.

The cylinder block is coupled with a cast oil pan to provide a lightweight, compact and very stiff bottom end. The cylinder heads are fitted with integrated intake manifold and camshaft covers. The exhaust manifolds are unique for each cylinder head. An acoustic cover is fitted over the upper engine to absorb engine-generated noise.

Depending on the market, there are 2 variants of the 3.0L V6 Diesel engine:

- Single Turbocharger variant
- Twin Turbocharger variant.

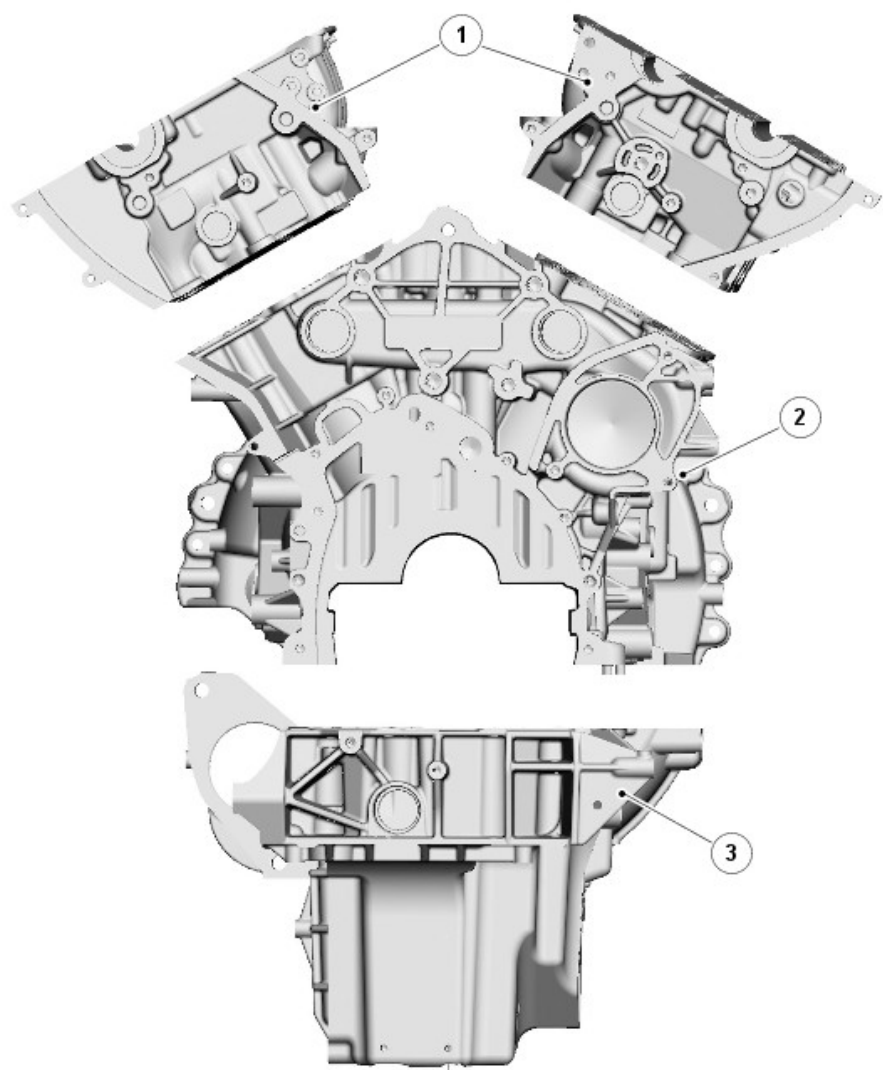
The single variant uses a single Variable Geometry Turbine (VGT) type turbocharger, the twin turbocharger variant uses two turbochargers. The twin turbocharger variant comprises a VGT type turbocharger- primary, and a fixed vane type (secondary turbocharger).

A low compression ratio of 16:1 contributes to improved emissions quality, quieter combustion and compatibility with the engine's unique forced induction system.

For additional information, refer to: [Intake Air Distribution and Filtering](#) (303-12A Intake Air Distribution and Filtering - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Description and Operation).

The low compression ratio also means less heat build-up in the piston bowl and more efficient fuel burn, resulting in the production of lower levels of pollutants. It also assists with cold starting allowing a faster cranking speed.

DESCRIPTION



E120983

ITEM		DESCRIPTION
1		Cylinder heads
2		Cylinder block
3		Oil pan

CYLINDER BLOCK COMPONENTS

The cylinder block is a single cast construction with a hollow beam structure. This type of construction provides outstanding strength and durability and reduces engine weight and length.

Lubrication oil is distributed through the cylinder block, via a main oil gallery

and channels bored in the block, to all critical moving parts. These channels divert oil to the main and connecting rod bearings via holes machined into the crankshaft.

A tapping at the rear of the cylinder block connects a pipe to the turbochargers by means of banjo connections. Oil is supplied, under pressure via this tapping, from the oil pump to provide lubrication for the bearings of the turbochargers.

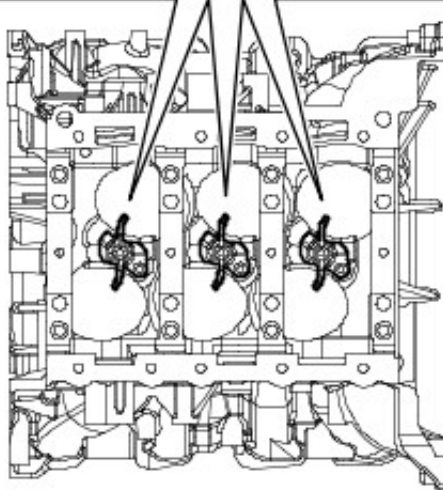
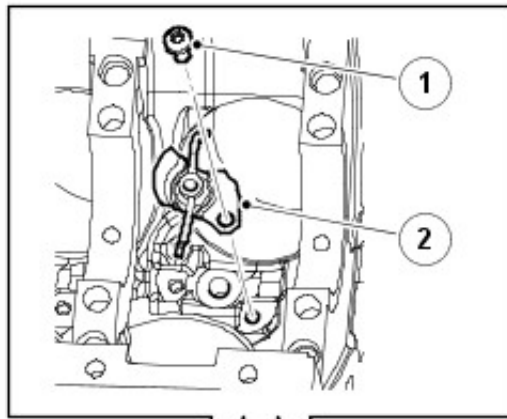
Cylinder cooling is achieved by coolant circulating through chambers in the cylinder block casting.

Two hollow dowels are used to locate the cylinder heads to the cylinder block, 1 on each side at the rear of the unit.

A port is included at the right and left side of the cylinder block, below each of the turbochargers, to connect the turbochargers oil return pipe to the oil pan.

Two coolant drain plugs are installed in the cylinder block; one is fitted in the rear right side, and the other is fitted in the middle of the cylinder block on the left side.

PISTON COOLING JETS

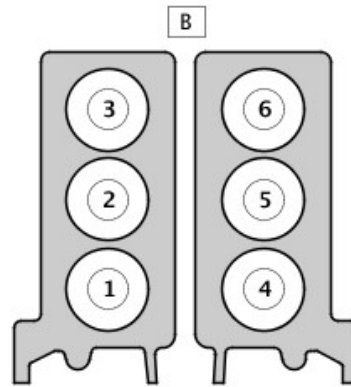


E44219

ITEM		DESCRIPTION
1		Bolt
2		Cooling jet

Jets located in the cylinder block provide piston and piston pin lubrication and cooling. These jets spray oil on to the inside of the piston, the oil then flows through 2 internal wave shaped channels to help cool each piston crown.

CYLINDER NUMBERING

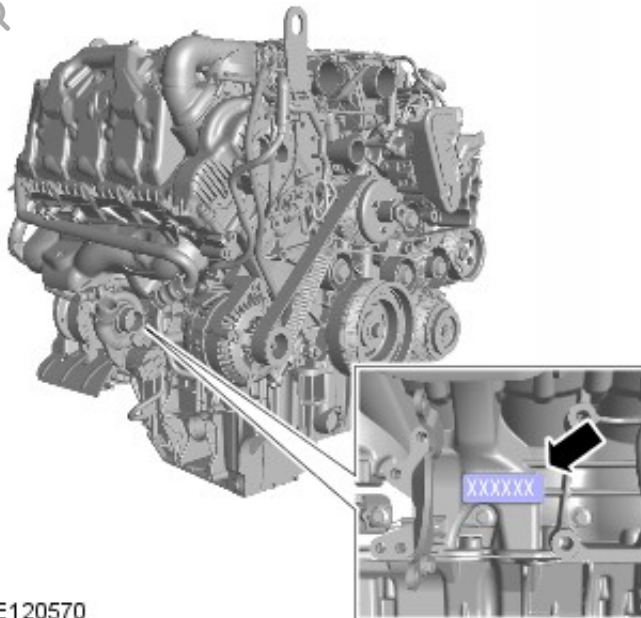


E163885

ITEM	DESCRIPTION
A	DIN standard cylinder numbering (A = engine rear)

DIN cylinder firing order 1,4,2,5,3,6

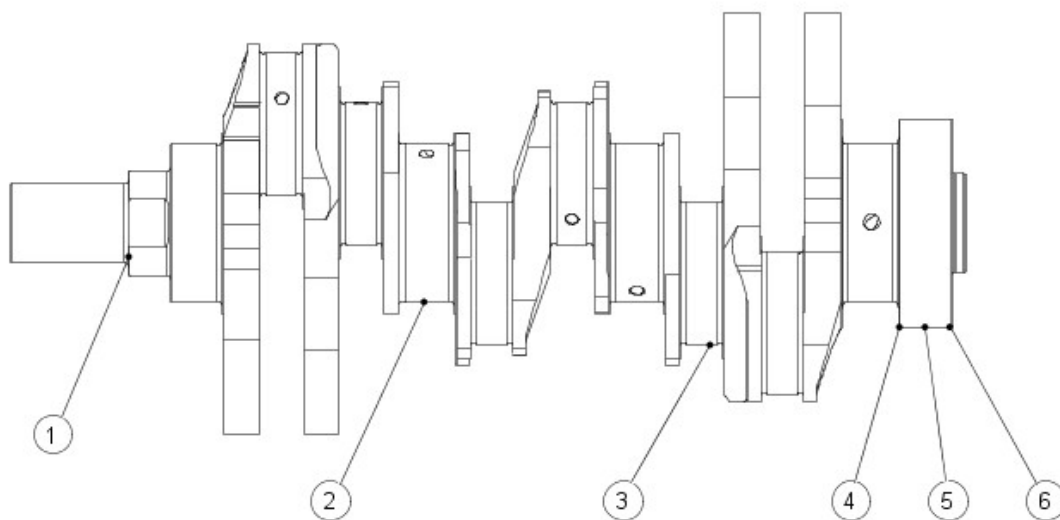
Engine Serial Number



E120570

Engine serial number is stamped on the right side of the cylinder block.

CRANKSHAFT



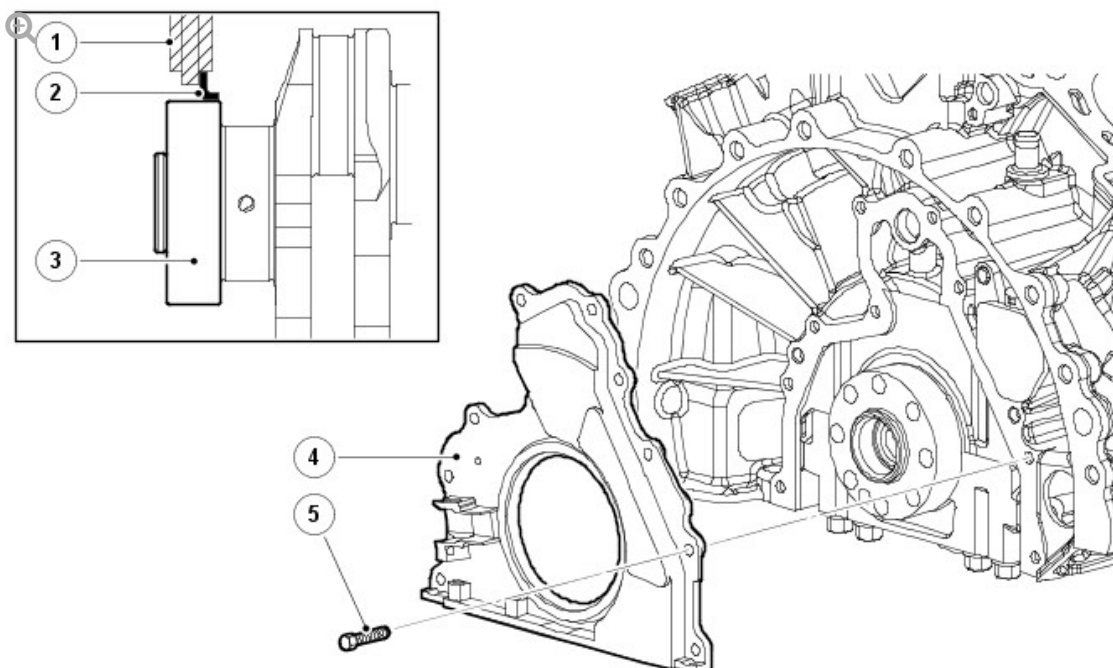
E52135

ITEM	DESCRIPTION
1	Oil pump drive
2	Main bearing journal
3	Connecting rod bearing journal
4	Rear drive flange
5	Rear oil seal location
6	Reluctor ring location

The crankshaft runs in 4 bearings with clamped 2 layer bearing shells. The upper and lower shells of bearing number 4 are flanged, which limits the end float of the crankshaft. The main bearing caps are double bolted and cross bolted to increase the strength and rigidity of the engine block.

The main bearings are selective split plain bearings. An oil groove in the top half of each bearing transfers oil into the crankshaft for lubrication of the connecting rod bearings.

CRANKSHAFT REAR OIL SEAL



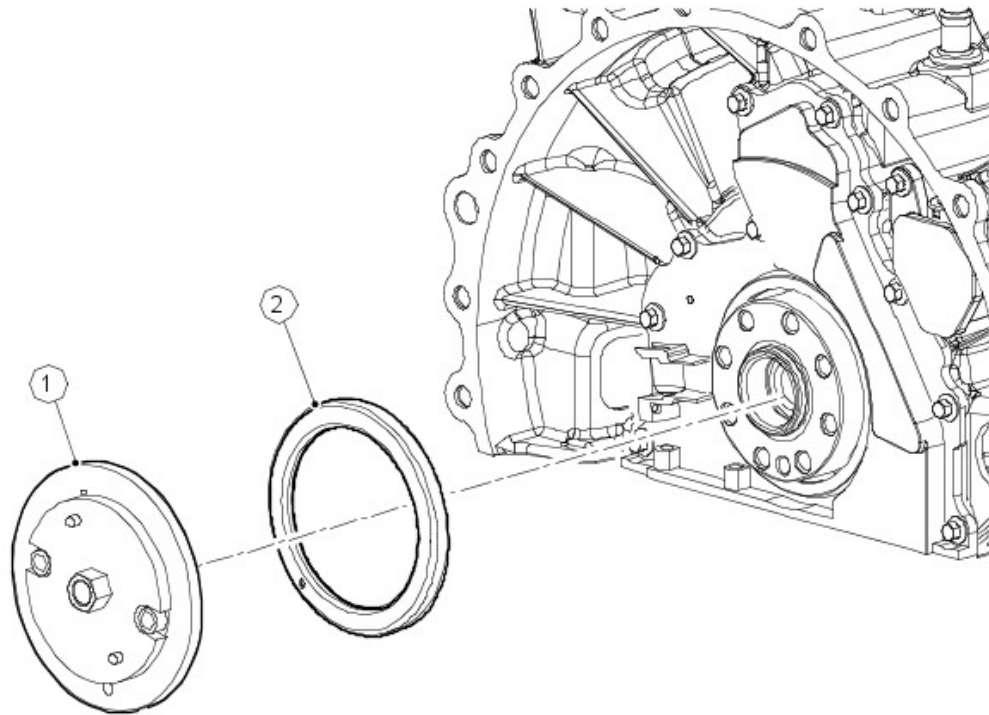
E44227

ITEM	DESCRIPTION
1	Housing
2	Seal
3	Crankshaft
4	Rear oil seal retainer
5	Bolt (10 off)

The rear main oil seal and retainer assembly is a one-piece unit and is supplied with its own fitting sleeve. The seal and retainer have 2 locating dowels, 10 fixing bolts and a seal. In addition, the retainer has a location for the Crankshaft Position Sensor (CKP).

For additional information, refer to: [Electronic Engine Controls](#) (303-14A Electronic Engine Controls - TDV6 3.0L Diesel, Description and Operation).

CRANKSHAFT RELUCTOR RING



E52137

ITEM		DESCRIPTION
1		Special tool
2		Reluctor ring

The crankshaft reluctor ring is located on the rear of the crankshaft. It is pressed onto the crankshaft using a special tool, which also precisely aligns the reluctor ring for crankshaft position and timing.

The reluctor ring consists of 60 magnets minus 2 for Engine Control Module (ECM) crankshaft position reference and synchronization. The magnets cannot be seen on the reluctor ring, which therefore can only be positioned using the special tool.

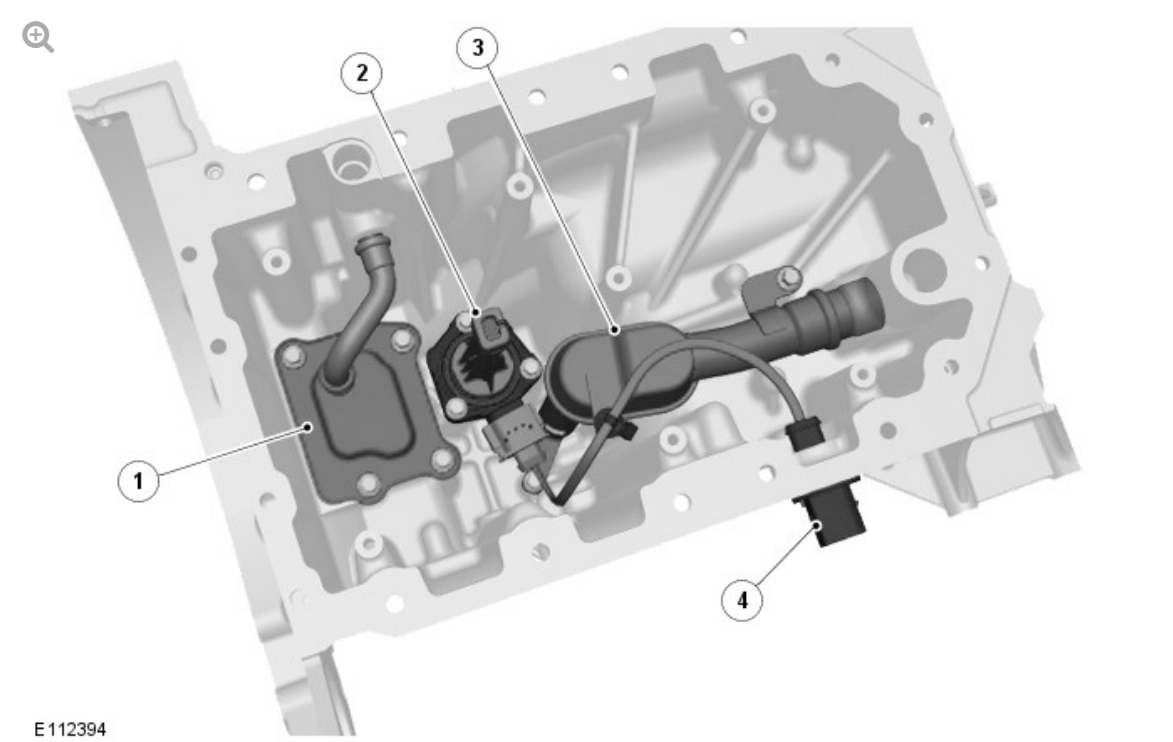
If the reluctor ring is removed for any reason, then a new reluctor ring must be fitted.

CRANKSHAFT DAMPER PULLEY

The crankshaft damper pulley is bolted to the front of the crankshaft to provide the drive for the engine accessory components.

For additional information, refer to: [Accessory Drive](#) (303-05A Accessory Drive - TDV6 3.0L Diesel, Description and Operation).

OIL PAN COMPONENTS



ITEM	DESCRIPTION
1	Oil scavenge reservoir
2	Oil level and temperature sensor
3	Oil pick-up pipe
4	Oil level and temperature sensor connector

The structural oil pan is fitted to the lower cylinder block to stiffen the base structure of the engine, helping to reduce Noise,Vibration and Harshness (NVH). The oil pan also incorporates an oil baffle plate to reduce oil foaming and splash.

The oil pan is secured to the cylinder block with 2 dowels, 2 locator pins for the gasket and 18 retaining bolts; 3 different lengths of bolts are used:

- M8 x 20 (9 off).

- M8 x 75 (1 off).
- M6 x 105 (8 off).

A gasket seals the joint between the oil pan and the cylinder block.

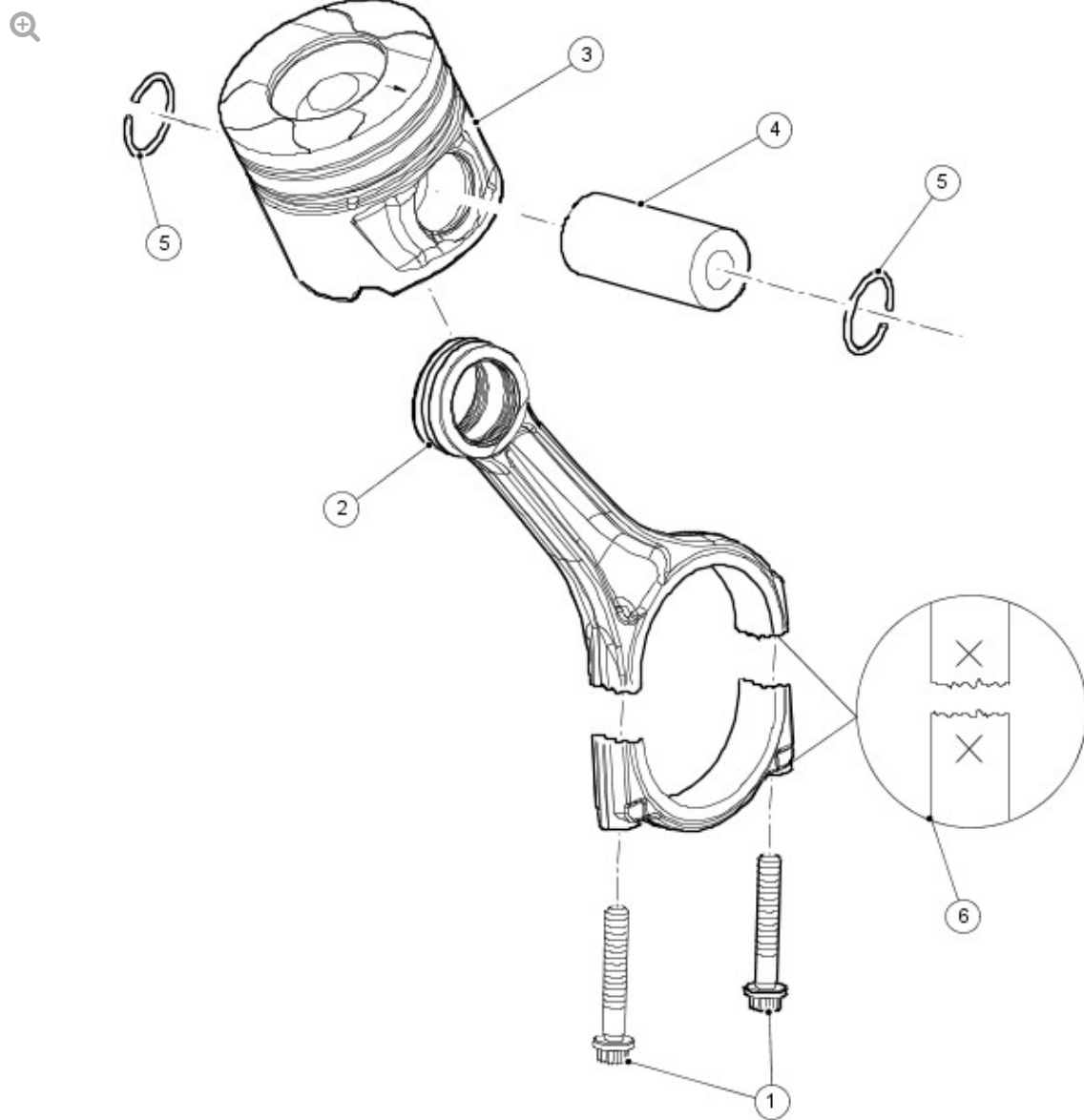
An oil pick-up pipe with integral strainer locates in the oil pan to provide oil to the crankshaft driven oil pump.

An oil level and temperature sensor provides an electronic indication of the oil level. This removes the requirement for the mechanical dipstick. An advantage in comparison with the static 'dipstick' method is that all marginal influences, for example vehicle being on a slope, lateral and longitudinal acceleration, are compensated for by averaging.

The values determined can be used to signal that the minimum oil level has been reached or to display the current oil level if required.

The sensor is mounted inside the oil pan where it sends an ultrasonic pulse vertically upward, it then measures the time for the pulse to be reflected back from the top surface of the oil.

PISTON AND CONNECTING ROD ASSEMBLY



E52134

ITEM	DESCRIPTION
1	Connecting rod bolts (2 off)
2	Connecting rod
3	Piston
4	Piston pin
5	Circlips
6	Connecting rod identification

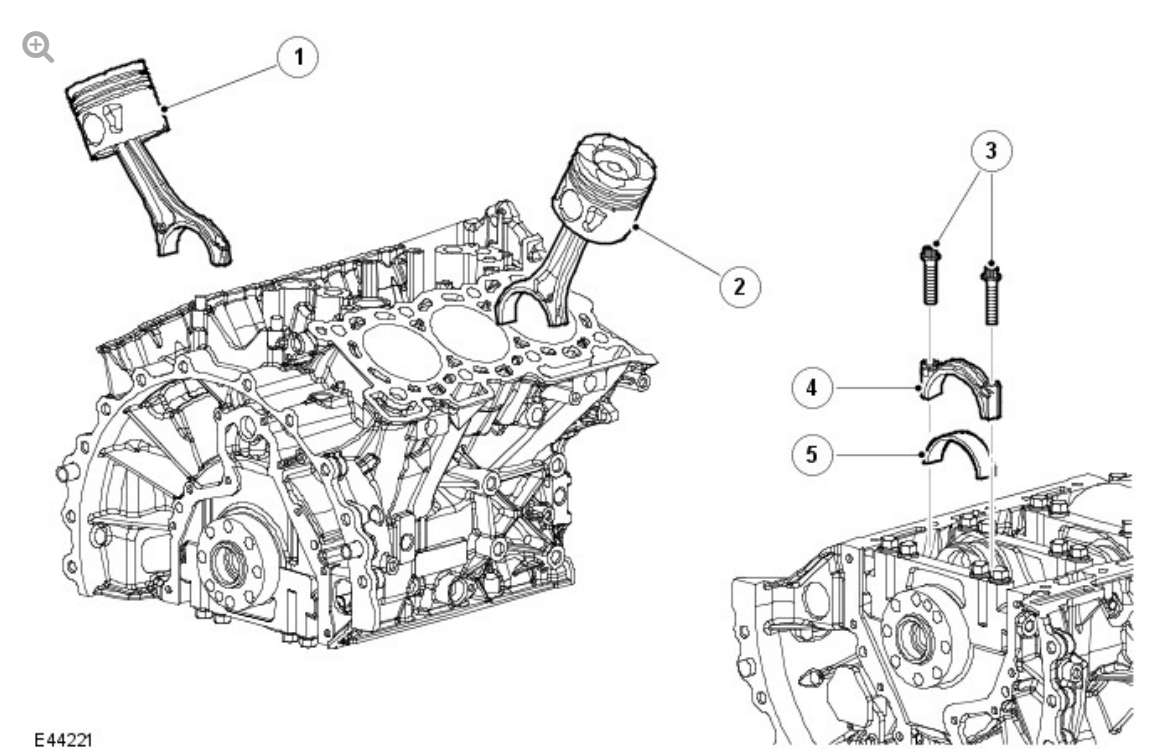
The connecting rods have fracture-split bearing caps. The bearing caps are produced by fracturing the opposing sides of the connecting rod at the

bearing horizontal center-line. As well as being easier to manufacture, when reassembled the fractured surfaces interlock to form a strong seamless joint. The cylinder position is etched on adjoining sides of the joint to identify matching connecting rods and bearing caps. The connecting rod bearings are selective split plain bearings. The connecting rod bearing is 'bismuth coated tri-metal', which is a manufacturing process that layers the bearing material to produce a higher load capacity for improved durability.

NOTE:

The connecting rods are not selective.

PISTON AND CONNECTING ROD ORIENTATION



E44221

ITEM	DESCRIPTION
1	Piston and connecting rod assembly, cylinders 4-6
2	Piston and connecting rod assembly, cylinders 1-3
3	Bolts (12 off)

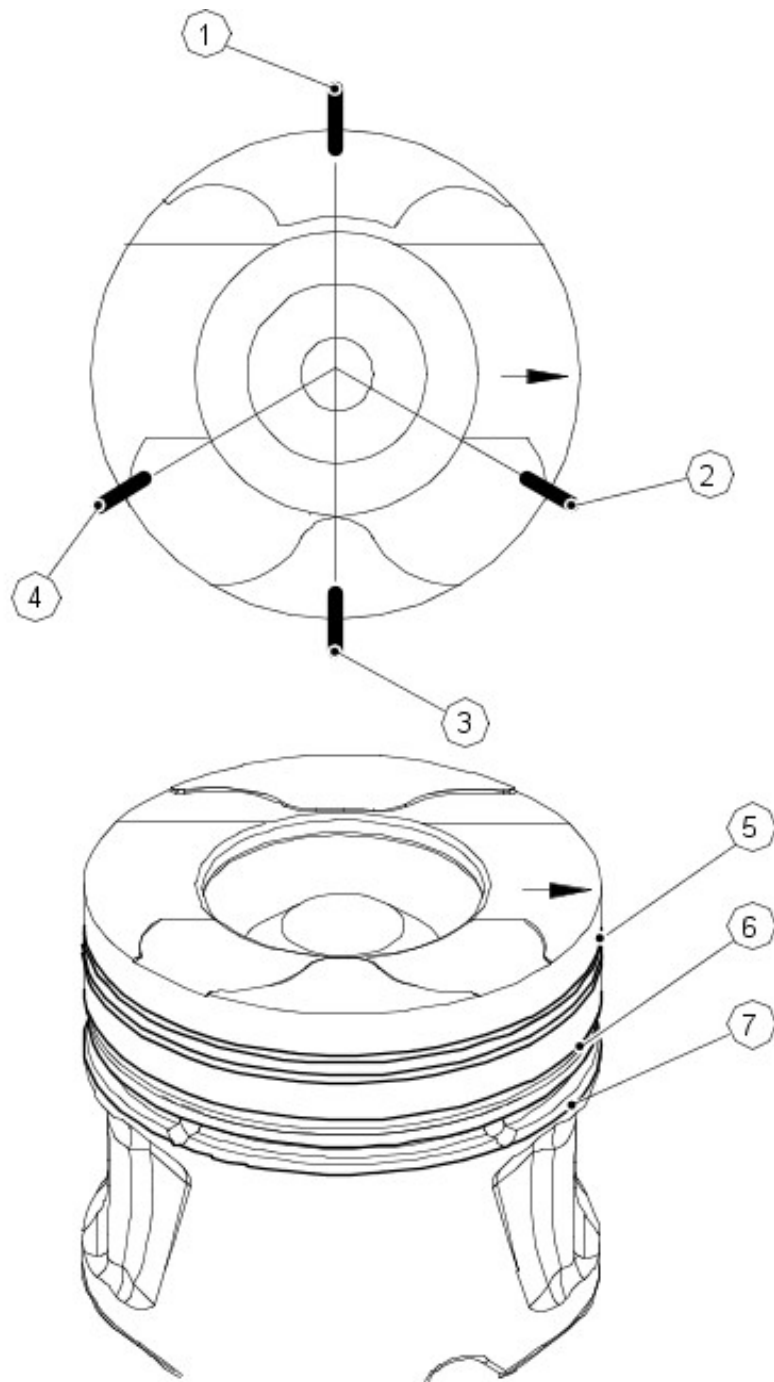
4	Connecting rod bearing cap (6 off)
5	Connecting rod lower bearing (6 off)

When installing a connecting rod, ensure the back of the connecting rod faces the center of the 'vee'.

The pistons are fitted with 3 rings. The piston crown incorporates a pronounced bowl; this forms the combustion chamber, which promotes swirl and turbulence necessary for good combustion and improved emissions. In addition, the piston skirt has a molybdenum-coated surface, which counteracts scoring of the cylinder bore and piston.

The piston also incorporates a double wave gallery within the piston crown to enhance piston cooling. The pistons are supplied oil by means of spray jets located in the cylinder block oil gallery. These jets ensure optimum piston cooling to counteract the high temperatures generated by the combustion process.

Each piston is installed on a piston pin located in a bushing in the connecting rod.



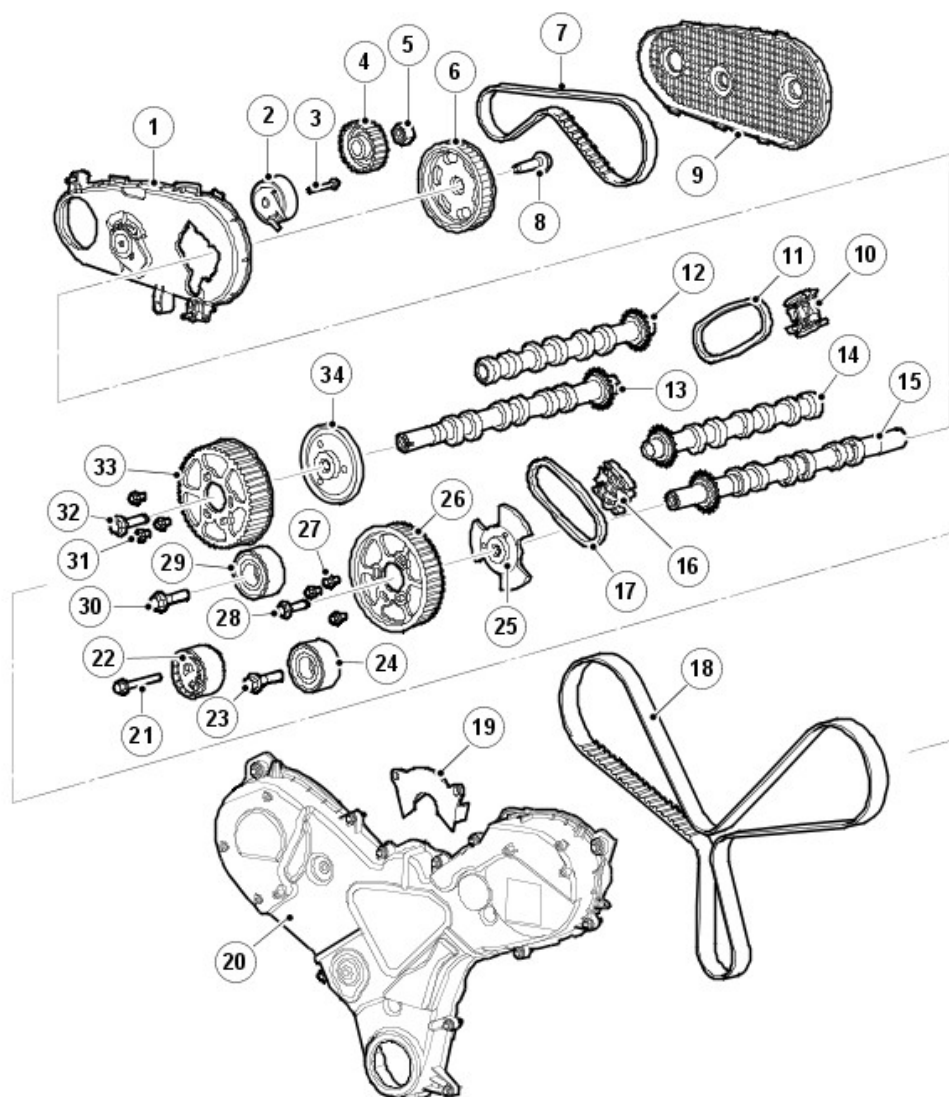
52133

ITEM	DESCRIPTION
1	Ring gap oil control
2	Ring gap upper compression
3	Spiral joint oil control
4	Ring gap lower compression
5	Upper compression ring
6	Lower compression ring

When installing pistons ensure the arrows on the piston crowns all point to the front of the engine. All pistons are common single grade/single part number for all engines.

The piston top ring is a taper type and is fitted with the taper to the top of the piston. All rings marked 'top' are assembled with 'top' uppermost. All rings must be spaced evenly around the piston before installing. The circumference gap of the double beveled oil control ring must be opposite the spiral control joint.

CAMSHAFT TIMING COMPONENTS



E94884

ITEM

DESCRIPTION

1	Timing belt front cover
2	Timing belt tensioner
3	Bolt
4	Fuel pump pulley
5	Nut
6	Timing belt pulley
7	Timing belt - High Pressure (HP) fuel pump
8	Bolt
9	Timing belt rear cover
10	Right chain tensioner
11	Right timing chain
12	Right intake camshaft
13	Right exhaust camshaft
14	Left intake camshaft
15	Left exhaust camshaft
16	Left chain tensioner
17	Left timing chain
18	Timing belt
19	Front cover bridge
20	Primary drive cover
21	Bolt
22	Tensioner
23	Bolt
24	Idler
25	Camshaft hub
26	Left camshaft timing pulley
27	Bolt (3 off)
28	Bolt

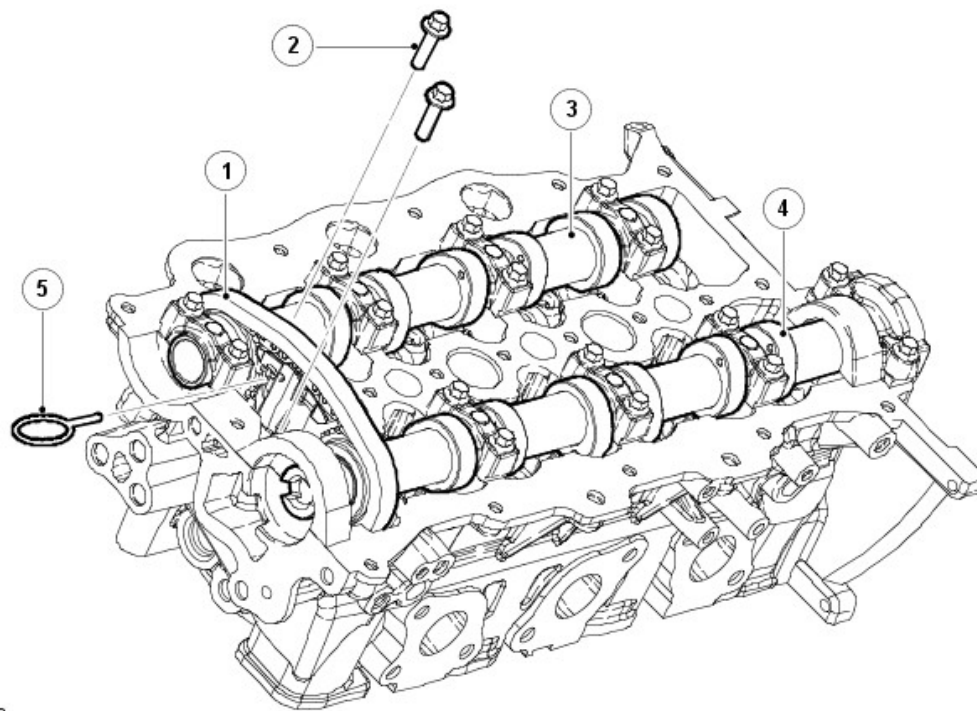
29	Idler
30	Bolt
31	Bolt (3 off)
32	Bolt
33	Right camshaft timing pulley
34	Camshaft hub

Primary drive is provided by a toothed timing belt from the crankshaft to the exhaust camshaft gears of each cylinder bank via 2 idler pulleys and a tensioner.

Timing belt adjustment is carried out by the eccentric type tensioner mounted on the right front face of the cylinder block.

The primary drive cover is secured to the front of the cylinder block, cylinder heads, intake manifolds and oil pump housing. The cover is sealed with a gasket. A bridge piece closes the back of the cover in between the cylinder heads.

SECONDARY DRIVE



E44233

ITEM	DESCRIPTION
1	Timing chain
2	Bolt (2 off)
3	Intake camshaft
4	Exhaust camshaft
5	Tensioner firing pin

Secondary drive is provided by 2 short crossover chains, which transfer drive from the exhaust camshaft gears to the intake camshaft gears. The crossover drives are located at the rear of the right cylinder bank and the front of the left cylinder bank. This allows for a much shorter and simpler run for the main camshaft drive belt at the front of the engine.

Each crossover chain is tensioned via an automatic chain tensioner, which acts directly on the chains via a guide rail. The tensioners are located between the exhaust and intake camshafts at the front or rear of the cylinder head, depending on the cylinder bank.

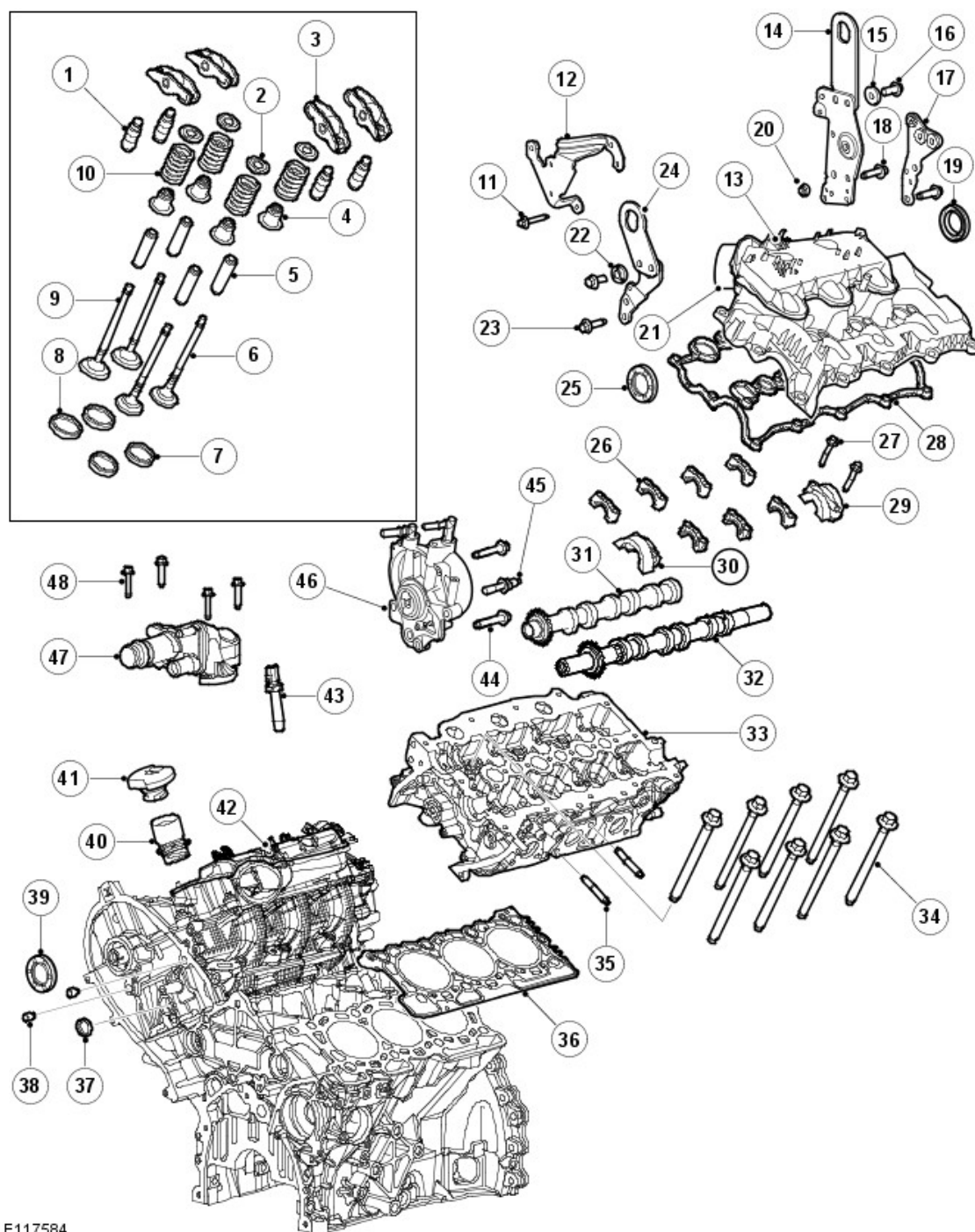
The tensioner firing pin holds the automatic chain tensioner in a

compressed state to aid installation.

CYLINDER HEAD COMPONENTS

NOTE:

Left cylinder head shown, right cylinder head similar.

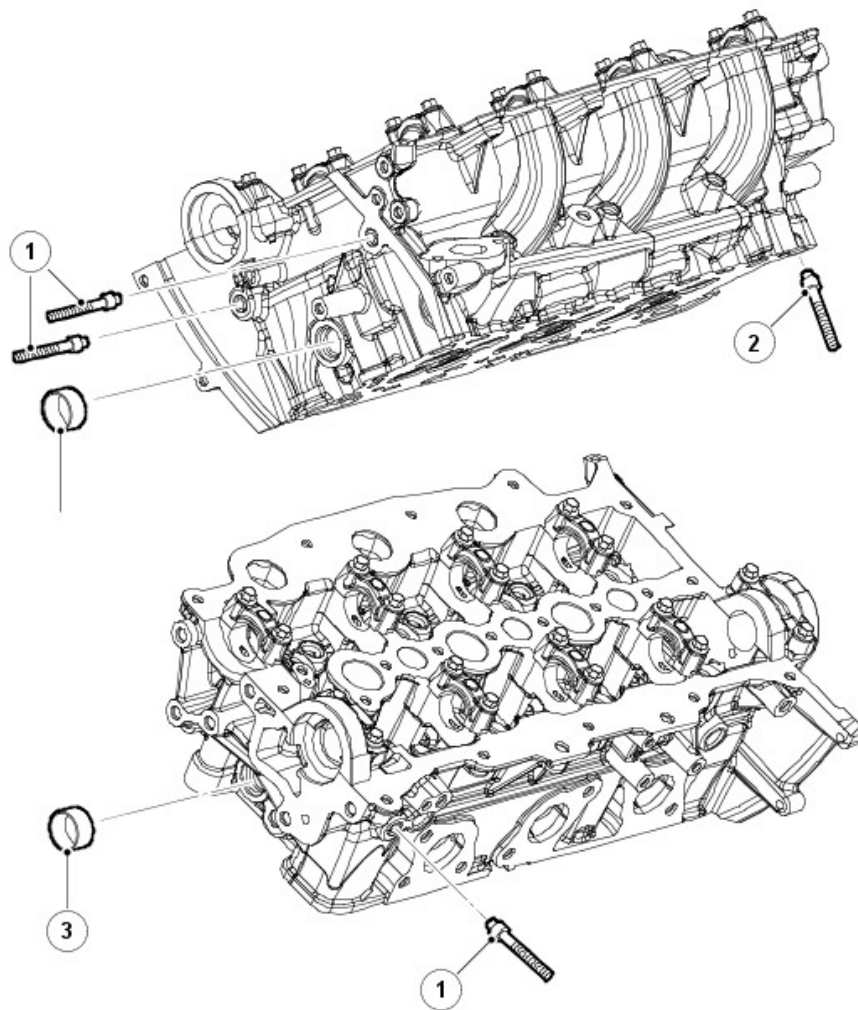


E117584

ITEM	DESCRIPTION
1	Hydraulic lash adjusters (12 off)
2	Valve spring retainers (12 off)
3	Rocker arms (12 off)
4	Valve stem seals (12 off)
5	Valve guides (12 off)
6	Exhaust valves (6 off)
7	Intake valve seats (6 off)
8	Exhaust valve seats (6 off)
9	Intake valves (6 off)
10	Valve springs (12 off)
11	Bolt
12	Bracket
13	Intake manifold
14	Lifting eye
15	Washer
16	Bolt
17	Bolt
18	Bolt
19	Seal
20	Cap
21	Intake manifold cover assembly
22	Washer
23	Bolt
24	Lifting eye
25	Seal
26	Camshaft bearing caps (7 off)

27	Bolts (18 off)
28	Gasket
29	Camshaft bearing cap and seal housing
30	Camshaft bearing cap and seal housing
31	Intake camshaft
32	Exhaust camshaft
33	Cylinder head
34	Cylinder head bolts (8 off)
35	Exhaust manifold studs (6 off)
36	Cylinder head gasket
37	Core plug
38	Plug
39	Seal
40	Oil filler tube
41	Oil filler cap
42	Intake manifold
43	Injectors (3 off)
44	Bolt (2 off)
45	Bolt
46	Vacuum and oil scavenge pump
47	Water outlet assembly
48	Bolt (4 off)

CYLINDER HEADS



E44236

ITEM		DESCRIPTION
1		Studs
2		Bolts
3		Blanking plug

The cylinder heads are unique to each cylinder bank. Eight deep-seated bolts help reduce distortion and secure each cylinder head to the cylinder block. The cylinder head bolts are located beneath the camshafts, 4 under the intake camshaft and 4 under the exhaust camshaft. Two hollow dowels align each cylinder head with the cylinder block.

NOTE:

The cylinder heads cannot be reworked.

The cylinder head gasket is available in 5 different thicknesses. The choice of gasket thickness is dependent on the maximum piston protrusion. Gasket thickness is identified by serrations cut into the front end of the gasket.

For additional information, refer to: [Specifications](#) (303-01A Engine - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Specifications).

The cylinder head has 4 ports machined at each cylinder location, 2 exhaust ports and 2 intake ports. One of the intake ports is helical and functions as a swirl port, the other is arranged laterally as a tangential port and functions as a charge port.

The camshafts are of a hollow tube construction, with pressed on lobes. Each camshaft is retained by caps, 5 for the exhaust camshafts and 4 for the intake camshafts. Location letters, A to I for the intake camshaft and R to Z for the exhaust camshaft, are marked on the outer faces of the caps for each cylinder head.

The left cylinder bank exhaust camshaft is machined to accept a rear camshaft gear. The rear camshaft gear provides drive for the high-pressure fuel pump.

For additional information, refer to: [Accessory Drive](#) (303-05A Accessory Drive - TDV6 3.0L Diesel, Description and Operation).

The exhaust camshaft gear of the left cylinder head also incorporates a reluctor ring, which is used in conjunction with the Camshaft Position Sensor (CMP) sensor to measure engine position.

For additional information, refer to: [Electronic Engine Controls](#) (303-14A Electronic Engine Controls - TDV6 3.0L Diesel, Description and Operation).

The right cylinder head exhaust camshaft is machined at the rear end to provide a drive connection for the vacuum and oil scavenge pump.

The fuel injection nozzles are centrally mounted above each cylinder.

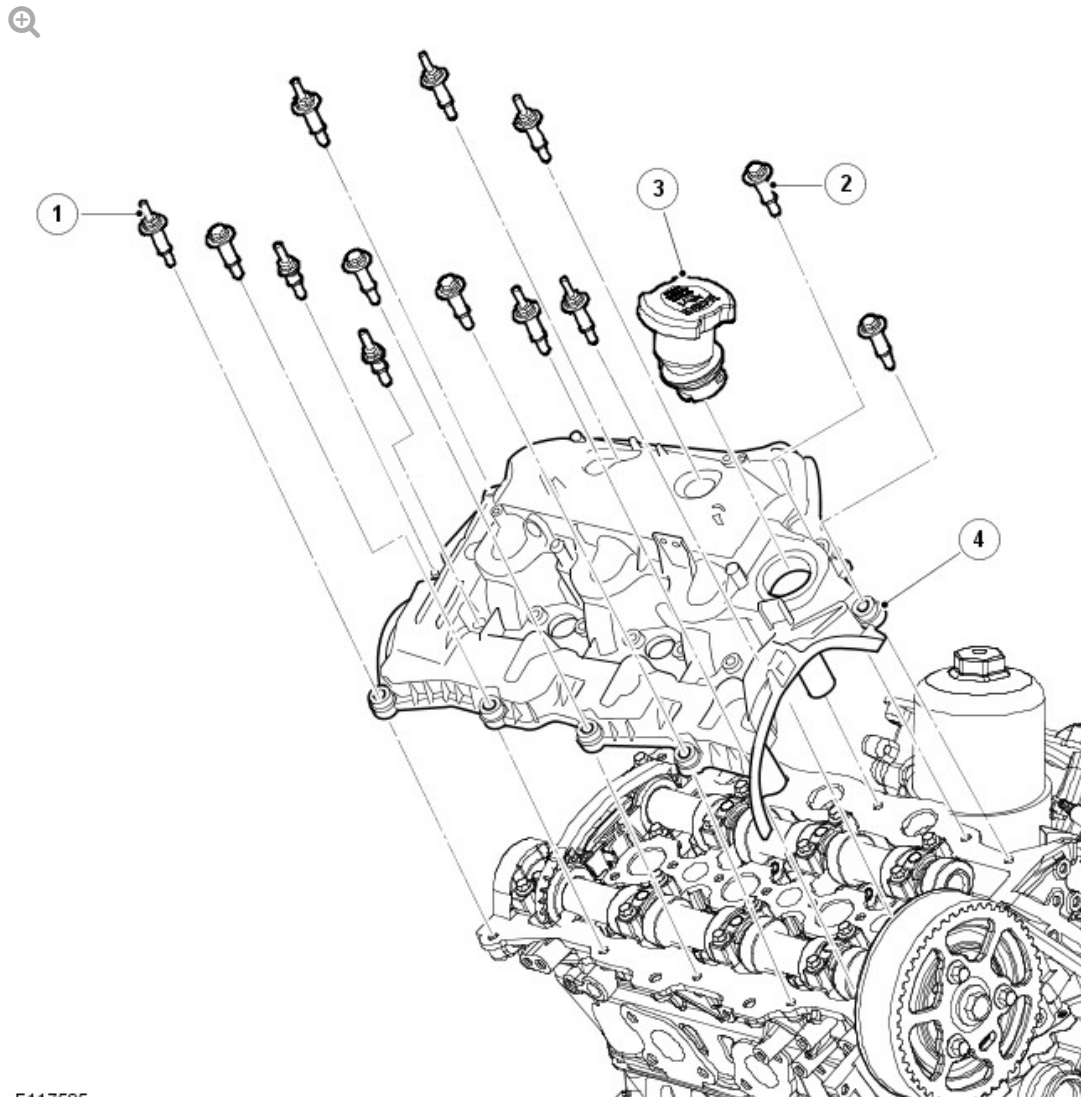
For additional information, refer to: [Fuel Charging and Controls](#) (303-04A Fuel Charging and Controls - TDV6 3.0L Diesel, Description and Operation).

The glow plugs are arranged centrally on the intake side of the cylinder heads, between the 2 intake ports of each cylinder.

For additional information, refer to: [Glow Plug System](#) (303-07C Glow Plug System - TDV6 3.0L Diesel, Description and Operation).

The engine lifting eyes are bolted to the cylinder head, 1 at the front and 2 at the rear, 1 per cylinder head.

CAMSHAFT COVERS



E117585

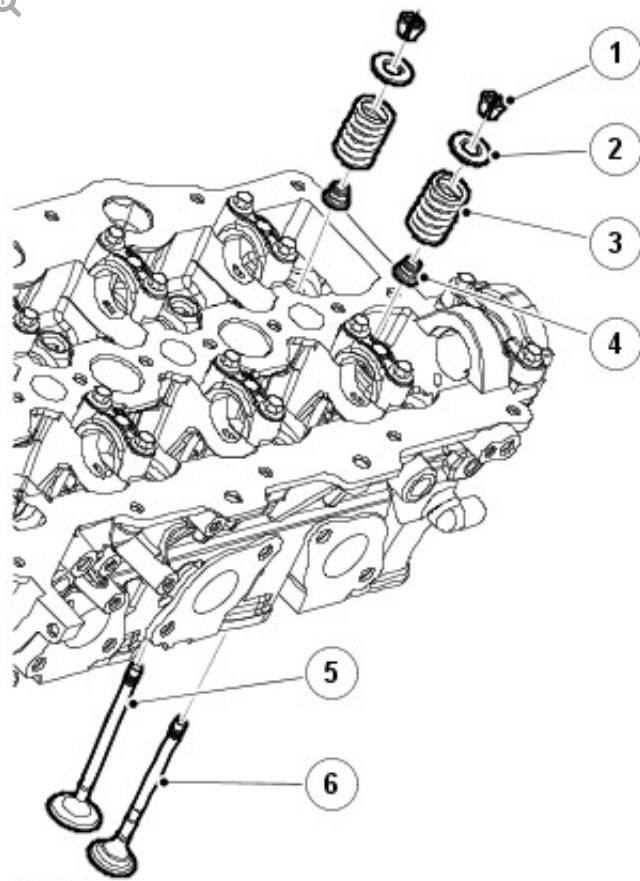
ITEM	DESCRIPTION
1	Stud bolt M6 x 40 (6 off)
2	Bolt M6 x 40 (7 off)
3	Oil filler cap
4	Right camshaft cover assembly

The right bank camshaft cover incorporates an outlet for the full load engine breather and the engine oil filler cap. The left bank camshaft cover incorporates an outlet for the part load engine breather.

For additional information, refer to: [Engine Emission Control](#) (303-08A Engine Emission Control - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Description and Operation).

In-groove gaskets seal the joints between the camshaft covers and the cylinder heads. Together with spacers and seals on the camshaft cover fasteners, they also isolate the covers from direct contact with the cylinder heads, to reduce noise.

INTAKE AND EXHAUST VALVES

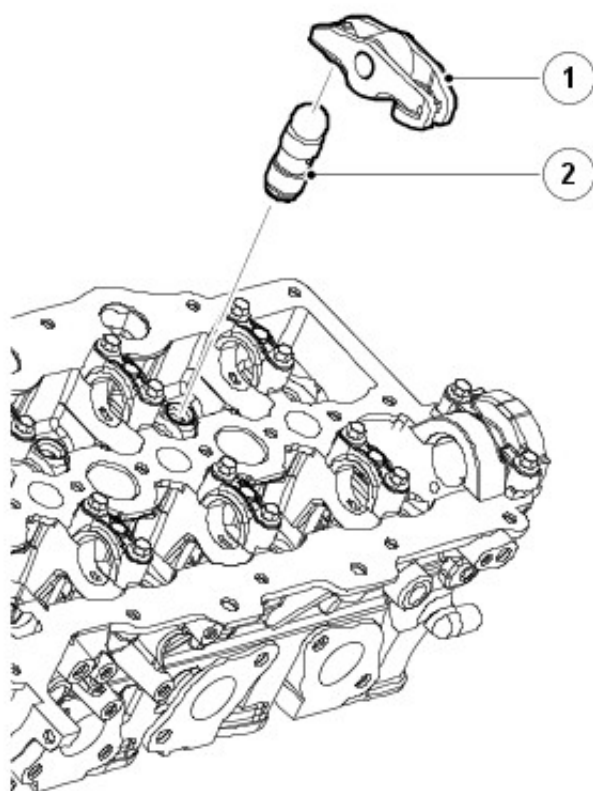


E44239

ITEM	DESCRIPTION
1	Valve spring collet (24 off, 12 per cylinder head)
2	Valve spring retainer (24 off, 12 per cylinder head)
3	Valve spring (24 off, 12 per cylinder head)
4	Valve stem seal (24 off, 12 per cylinder head)
5	Intake valve (12 off, 6 per cylinder head)
6	Exhaust valve (12 off, 6 per cylinder head)

Each cylinder head incorporates 2 overhead camshafts operating 4 valves per cylinder via rocker arms with hydraulic lash adjusters.

ROCKER ARMS AND HYDRAULIC LASH ADJUSTERS

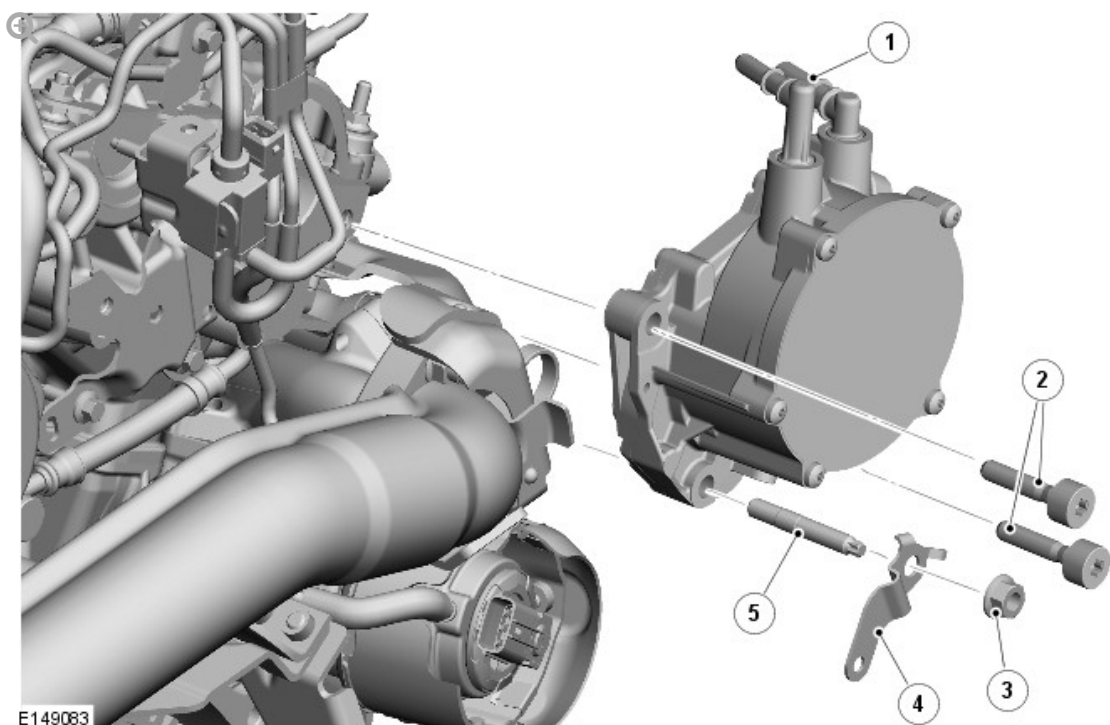


E44238

ITEM	DESCRIPTION
1	Rocker arm (24 off, 12 per cylinder head)
2	Hydraulic lash adjuster (24 off, 12 per cylinder head)

The lightweight valve gear provides good economy and noise levels. Valve head diameters are 31 mm (1.220 in) for the exhaust and 35 mm (1.378 in) for the intake. All valves have 5 mm (0.197 in) diameter stems supported in seats and guide inserts. Valve spring collets, collars seats locate single valve springs on both intake and exhaust valves. Valve stem seals are integrated into the valve spring seats.

VACUUM AND OIL SCAVENGE PUMP



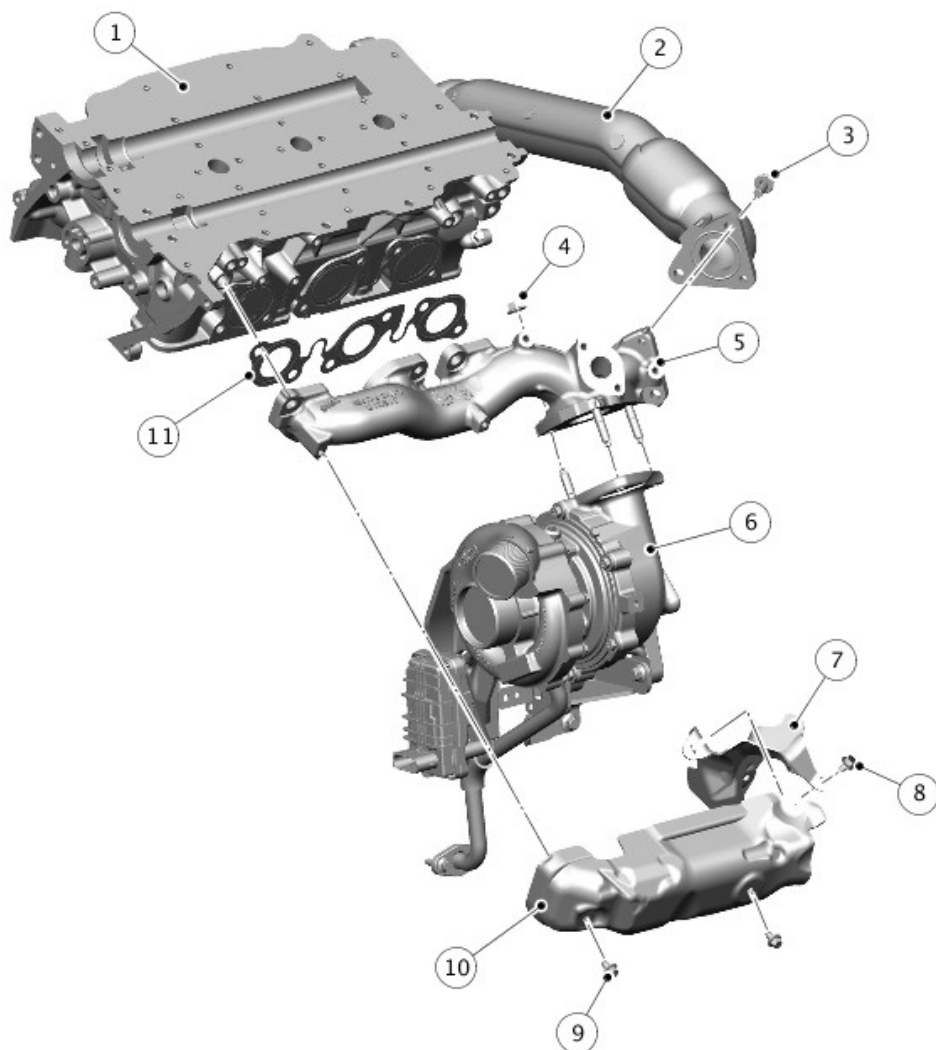
ITEM	DESCRIPTION
1	Vacuum pipe connections
2	Torx bolt (2 off)
3	Nut
4	Bracket (emission hose)
5	Stud

The vacuum and oil scavenge pump is located at the rear of the right cylinder head and driven from the exhaust camshaft. The pump provides the vacuum used by the Exhaust Gas Recirculation (EGR), turbocharger and brake booster systems, and scavenges oil from the turbochargers.

EXHAUST MANIFOLDS

NOTE:

Left exhaust manifold is shown, right exhaust manifold is similar.



E169819

ITEM	DESCRIPTION
1	Cylinder head
2	Connecting pipe
3	Bolt (3 off)
4	Nut (3 off)
5	Exhaust manifold
6	Turbocharger (primary turbocharger shown)
7	Manifold rear heat shield
8	Bolt
9	Bolt (2 off)
10	Manifold heat shield

The exhaust manifolds are sealed to the cylinder head by means of a gasket. Sacrificial sleeves are used to align the manifolds. These sleeves must be changed when refitting the manifolds.

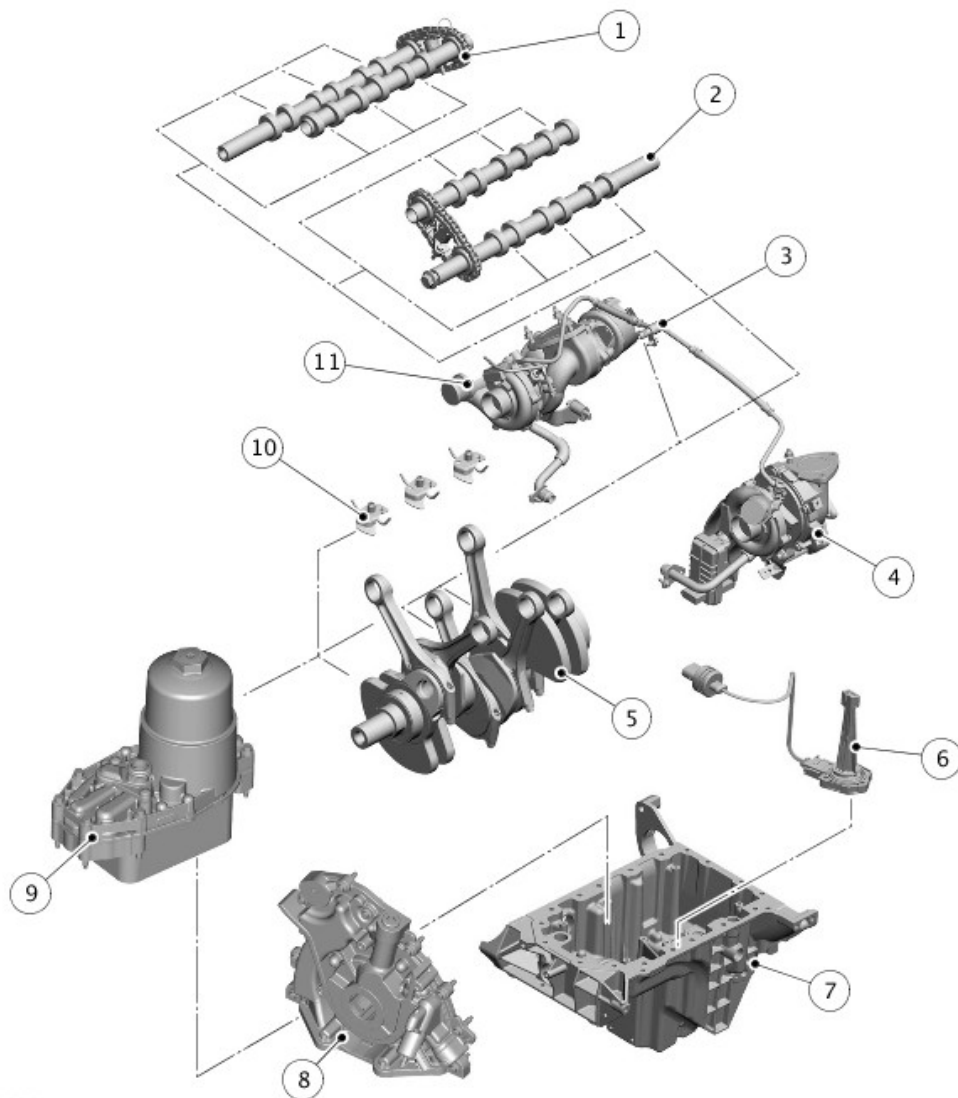
Each manifold has a connection for the EGR transfer pipe.

The engine is fitted with twin variable geometry turbochargers, which fix to the exhaust manifolds by a 3-hole flange and a gasket.

LUBRICATION SYSTEM

NOTE:

Twin Turbocharger version shown, Monoturbo version similar.



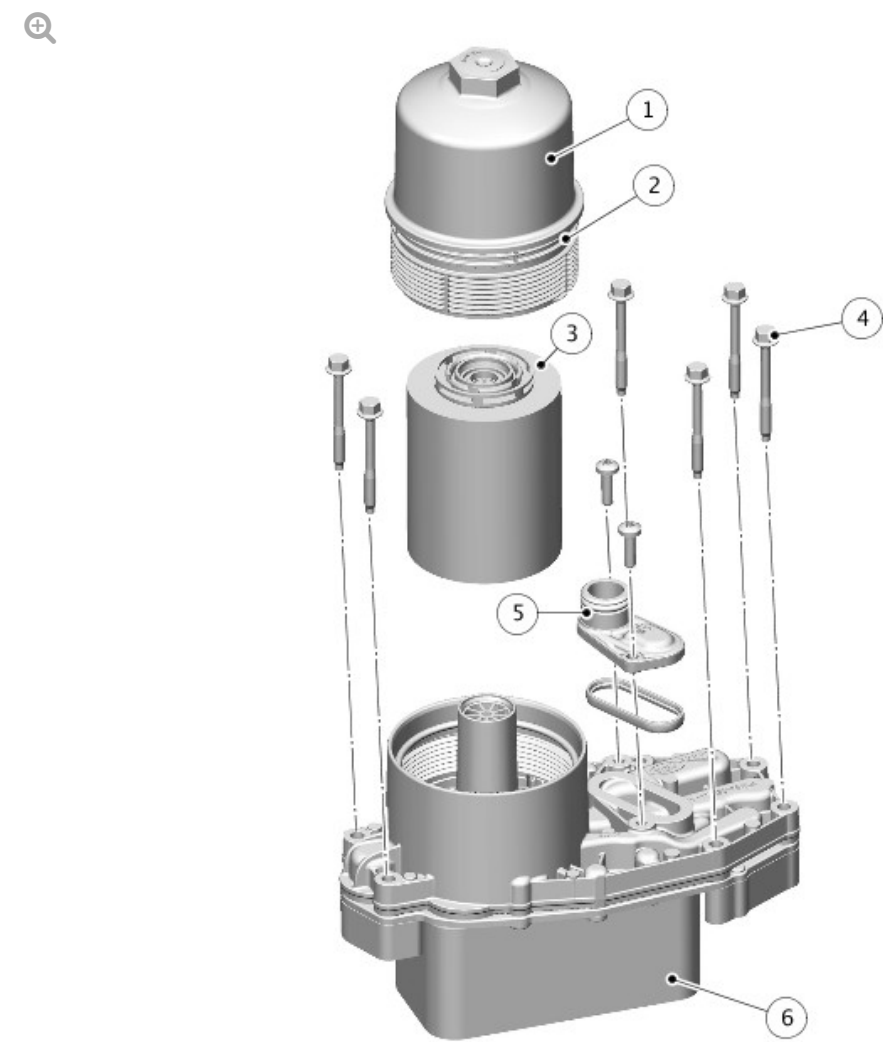
E169820

ITEM	DESCRIPTION
1	Intake camshaft
2	Exhaust camshaft
3	Turbocharger oil supply
4	Primary turbocharger
5	Crankshaft and connecting rods
6	Oil level and temperature sensor
7	Oil pan
8	Oil pump
9	Oil cooler and filter assembly
10	Piston cooling jets

Oil is drawn from the oil pan and pressurized by the oil pump. The output from the oil pump is then filtered and distributed through internal oil passageways.

All moving parts are lubricated by pressure or splash oil. Pressurized oil is also provided for operation of the hydraulic lash adjusters and the timing gear chain tensioners.

OIL COOLER AND FILTER ASSEMBLY



E169821

ITEM		DESCRIPTION	
1		Filter housing	

2	O-ring seal
3	Paper element
4	Retaining bolt (6 off)
5	Coolant outlet connection
6	Cooler assembly

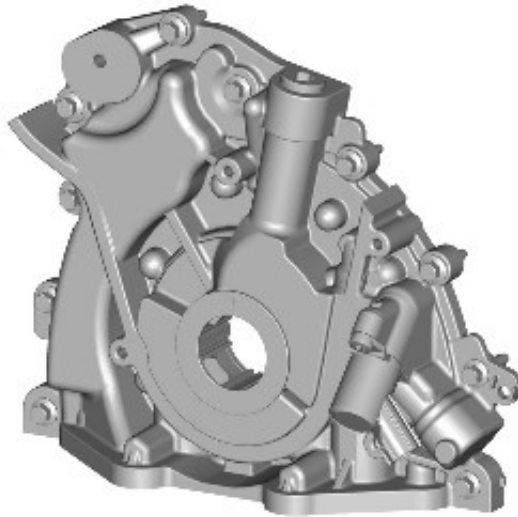
The engine is lubricated by a force-feed oil circulation system with a full flow oil filter. The oil cooler forms a unit with the oil filter, which is mounted centrally in the middle of the cylinder block between the 2 banks of cylinders. The engine oil is cooled using the engine cooling system.

Oil returns to the oil pan under gravity. Large drain holes through the cylinder heads and cylinder block ensure the quick return of the oil, reducing the volume of oil required and enabling an accurate check of the contents soon after the engine stops.

System replenishment is through the oil filler cap on the right camshaft cover.

The oil pick-up is immersed in the oil reservoir to provide a supply to the oil pump during all normal vehicle attitudes. A mesh screen in the intake prevents debris from entering the oil system.

OIL PUMP



E169823

The oil pump is a G-rotor 2-stage pressure pump with solenoid control to switch between low and high pressure modes depending on engine speed and load to optimize fuel economy.

It is bolted to the front of the cylinder block. It is sealed by means of a gasket, which is recessed into the oil pump housing. The pump intake and outlet ports align with oil passages in the cylinder block.

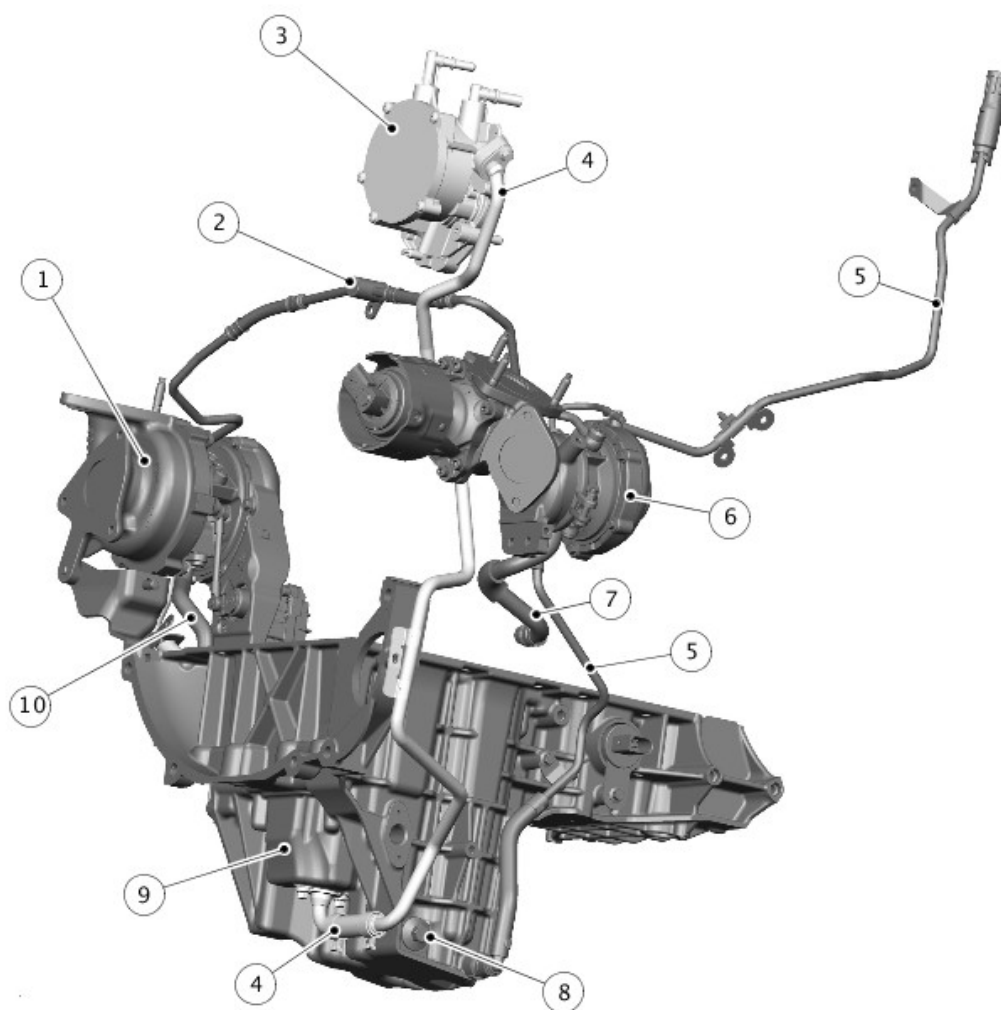
The pumping element is an eccentric rotor, which is directly driven by flats on the crankshaft. An integral pressure relief valve regulates pump outlet pressure at 4.5 bar (65.25 lb/in²).

The front crankshaft oil seal is housed in the oil pump casing and is fitted such that its front face is 1 mm (0.04 in) under flush with the machined front face of the oil pump.

NOTE:

The seal is not to be pushed all the way into the bore as this will block the seal drains.

OIL PAN



E169822

ITEM	DESCRIPTION
1	Primary turbocharger
2	Turbocharger oil feed pipes
3	Vacuum and oil scavenge pump
4	Oil scavenge pipe
5	Oil extraction tube
6	Secondary Turbocharger
7	Secondary Turbocharger oil return tube
8	Drain plug
9	Secondary reservoir
10	Primary Turbocharger oil return tube

Due to the wide range of inclines Land Rover vehicles operate across, the geometry of the oil pan has been configured to guarantee oil pick-up across all operating angles. An oil scavenge system has been developed to guarantee excellent oil flow through the turbochargers on severe side-slopes. At extreme angles there is a risk the turbochargers may be below the oil level in the oil pan, restricting oil return flow.

To overcome this, the engine uses:

- The oil pump to supply oil to the turbocharger bearings from the main reservoir of the oil pan.
- A secondary reservoir in the oil pan, sealed off from the main reservoir, to receive oil from the turbocharger bearings.
- The vacuum and oil scavenge pump to assist the oil flow from the turbocharger bearings by scavenging oil from the secondary reservoir into the right camshaft cover.

For service, a drain plug and an oil extraction tube are installed to drain the main body of the oil pan.

OIL LEVEL CHECK

The engine oil level can be displayed in the message center on the Instrument Cluster.

For accuracy, oil level checks should be performed with the vehicle on level ground when the oil is hot. The vehicle needs to stand for approximately 10 minutes, after the engine is switched off, to allow the oil to drain back into the sump and the oil level to stabilize. The oil level system will not give a reading until the oil level has stabilized.

The method of checking the oil level can be found in the oil draining and filling procedures. For additional information, refer to: (303-01F Engine - SDV6 3.0L Diesel - Hybrid Electric Vehicle)

[Engine Oil Draining and Filling](#) (General Procedures),
[Engine Oil Vacuum Draining and Filling](#) (General Procedures).

OPERATION

Operation of the engine is controlled by the ECM.

For additional information, refer to: [Electronic Engine Controls](#) (303-14A
Electronic Engine Controls - TDV6 3.0L Diesel, Description and Operation).